ADVANCED TOPICS

Dimension Reduction and Latent Variables

From Canonical Correlation Analysis to Variational Autoencoders

Beyond PCA: The Need for Advanced Methods

RECAP: THE LIMITS OF PCA

Principal Component Analysis (PCA) is the foundational tool for dimensionality reduction, but it operates under strict assumptions:

- **Linearity:** PCA only captures linear correlations. It fails to unfold complex, non-linear manifolds (e.g., the "Swiss Roll").
- **Single Modality:** PCA is designed for a single dataset $\mathbf{X}$. It cannot naturally find shared structures across multiple views (e.g., genomics + imaging).
- **Deterministic Mapping:** Standard PCA provides a deterministic projection, lacking a probabilistic generative mechanism for sampling new data points.

THE ADVANCED LANDSCAPE

To address these limitations, we move to more sophisticated latent variable models:

- **Multi-Modal Analysis:** Canonical Correlation Analysis (CCA) finds shared linear subspaces across multiple datasets.
- **Non-Linear Manifolds:** t-SNE and UMAP preserve local (and global) topological structures for visualization.
- **Neural & Generative Models:** Autoencoders learn non-linear mappings, while Variational Autoencoders (VAEs) provide a rigorous probabilistic framework for generation.

> **Lecture Objective:** To understand the mathematical foundations of these advanced techniques, their optimization objectives, and when to apply them in high-dimensional research.

Canonical Correlation Analysis (CCA): Multi-Modal PCA

THE MULTI-MODAL PROBLEM

In many modern applications, we have multiple "views" or modalities of the same underlying entities. For example:

- **Genomics:** Gene expression ($\mathbf{X}$) and DNA methylation ($\mathbf{Y}$) for the same patients.
- **Multimedia:** Images ($\mathbf{X}$) and their corresponding text descriptions ($\mathbf{Y}$).

Goal: Find a shared latent space that captures the common information between these modalities, ignoring modality-specific noise.

> **Analogy to PCA:** While PCA finds linear combinations that maximize *variance* within a single dataset, CCA finds linear combinations that maximize *correlation* between two datasets.

MATHEMATICAL SETUP

Given two centered datasets $\mathbf{X} \in \mathbb{R}^{N \times p}$ and $\mathbf{Y} \in \mathbb{R}^{N \times q}$ measured on the same N samples.

We seek projection vectors $\mathbf{a} \in \mathbb{R}^p$ and $\mathbf{b} \in \mathbb{R}^q$ such that the projected variables $\mathbf{u} = \mathbf{X}\mathbf{a}$ and $\mathbf{v} = \mathbf{Y}\mathbf{b}$ are maximally correlated.

THE OBJECTIVE FUNCTION

The empirical correlation between $\mathbf{u}$ and $\mathbf{v}$ is:

$$\rho = \frac{\mathbf{a}^\top \mathbf{\Sigma}_{XY} \mathbf{b}}{\sqrt{\mathbf{a}^\top \mathbf{\Sigma}_{XX} \mathbf{a}} \sqrt{\mathbf{b}^\top \mathbf{\Sigma}_{YY} \mathbf{b}}}$$

Where $\mathbf{\Sigma}_{XX}$, $\mathbf{\Sigma}_{YY}$ are auto-covariance matrices, and $\mathbf{\Sigma}_{XY}$ is the cross-covariance matrix.

CCA: Mathematical Formulation and Solution

CONSTRAINED OPTIMIZATION

The correlation objective is invariant to scaling of $\mathbf{a}$ and $\mathbf{b}$. We can fix the scale by constraining the variance of the projected variables to be 1:

$$\max_{\mathbf{a},\mathbf{b}} \mathbf{a}^\top \mathbf{\Sigma}_{XY} \mathbf{b}$$

Subject to the constraints:

$$\mathbf{a}^\top \mathbf{\Sigma}_{XX} \mathbf{a} = 1 \quad \text{and} \quad \mathbf{b}^\top \mathbf{\Sigma}_{YY} \mathbf{b} = 1$$

Using Lagrange multipliers λ_1 and λ_2, the Lagrangian is:

$$\mathcal{L} = \mathbf{a}^\top \mathbf{\Sigma}_{XY} \mathbf{b} - \frac{\lambda_1}{2}(\mathbf{a}^\top \mathbf{\Sigma}_{XX} \mathbf{a} - 1) - \frac{\lambda_2}{2}(\mathbf{b}^\top \mathbf{\Sigma}_{YY} \mathbf{b} - 1)$$

GENERALIZED EIGENVALUE PROBLEM

Taking derivatives with respect to $\mathbf{a}$ and $\mathbf{b}$ and setting them to zero yields the system:

$$\mathbf{\Sigma}_{XY} \mathbf{b} = \lambda_1 \mathbf{\Sigma}_{XX} \mathbf{a} \mathbf{\Sigma}_{YX} \mathbf{a} = \lambda_2 \mathbf{\Sigma}_{YY} \mathbf{b}$$

Multiplying the first by $\mathbf{a}^\top$ and the second by $\mathbf{b}^\top$ shows that $\lambda_1 = \lambda_2 = \lambda$, which is the correlation ρ.

THE SOLUTION

Substituting $\mathbf{b}$ from the second equation into the first gives a generalized eigenvalue problem for $\mathbf{a}$:

$$\mathbf{\Sigma}_{XX}^{-1} \mathbf{\Sigma}_{XY} \mathbf{\Sigma}_{YY}^{-1} \mathbf{\Sigma}_{YX} \mathbf{a} = \lambda^2 \mathbf{a}$$

Result: The optimal projection $\mathbf{a}$ is the leading eigenvector; the maximum correlation is λ.

Non-linear Dimensionality Reduction: The Crowding Problem

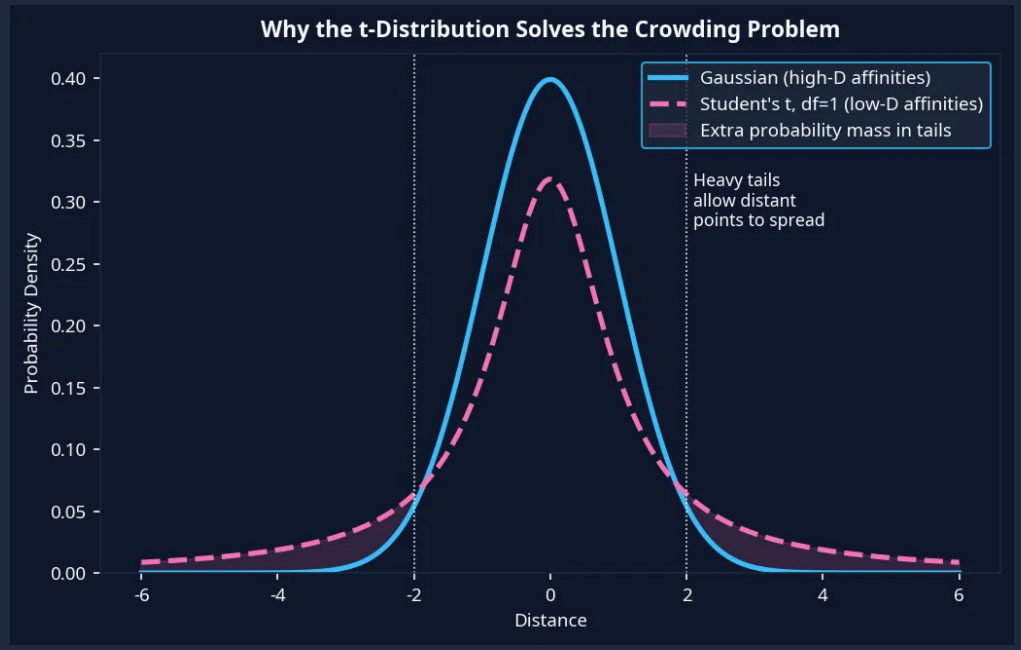
THE "CROWDING PROBLEM"

In high-dimensional space (e.g., $D = 100$), the volume of a sphere scales as r^D. There is vastly more "room" at a given distance r than in a low-dimensional space (e.g., $d = 2$).

When embedding into 2D, we cannot faithfully represent all pairwise distances. If we accurately model local neighborhoods, moderately distant points are forced too close together, causing all points to collapse into a dense center.

The Solution (t-SNE): Use a heavy-tailed distribution (Student's t-distribution with 1 degree of freedom, i.e., a Cauchy distribution) in the low-dimensional space.

This allows moderately distant points in high-D to be placed much further apart in low-D without incurring a massive penalty, effectively solving the crowding problem.



t-SNE: Mathematical Formulation

HIGH-DIMENSIONAL SPACE

In the original space $\mathbb{R}^D$, we define the similarity between points $\mathbf{x}_i$ and $\mathbf{x}_j$ as the conditional probability $p_{j|i}$ that $\mathbf{x}_i$ would pick $\mathbf{x}_j$ as its neighbor if neighbors were picked in proportion to their probability density under a Gaussian centered at $\mathbf{x}_i$:

$$p_{j|i} = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2/2\sigma_i^2)}{\sum_{k\neq i}\exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2/2\sigma_i^2)}$$

The variance σ_i^2 is chosen such that the perplexity of the conditional distribution matches a user-specified value.

To simplify optimization, we define a symmetric joint probability:

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}$$

LOW-DIMENSIONAL SPACE

In the target space $\mathbb{R}^d$ (usually $d = 2$ or 3), we define the similarity q_{ij} between the corresponding points $\mathbf{y}_i$ and $\mathbf{y}_j$.

Instead of a Gaussian, t-SNE uses a **Student's t-distribution with one degree of freedom** (a Cauchy distribution) to measure similarities in the low-dimensional space:

$$q_{ij} = \frac{(1 + \|\mathbf{y}_i - \mathbf{y}_j\|^2)^{-1}}{\sum_{k\neq l}(1 + \|\mathbf{y}_k - \mathbf{y}_l\|^2)^{-1}}$$

Why the t-distribution? The heavy tails of the t-distribution allow moderately distant points in the high-dimensional space to be modeled by points that are much further apart in the low-dimensional map, effectively solving the **crowding problem**.

t-SNE: Optimization via KL Divergence

THE OBJECTIVE FUNCTION

t-SNE minimizes the **Kullback-Leibler (KL) divergence** between the high-dimensional probabilities p_{ij} and the low-dimensional probabilities q_{ij}:

$$C = KL(P||Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

Asymmetry of KL Divergence:

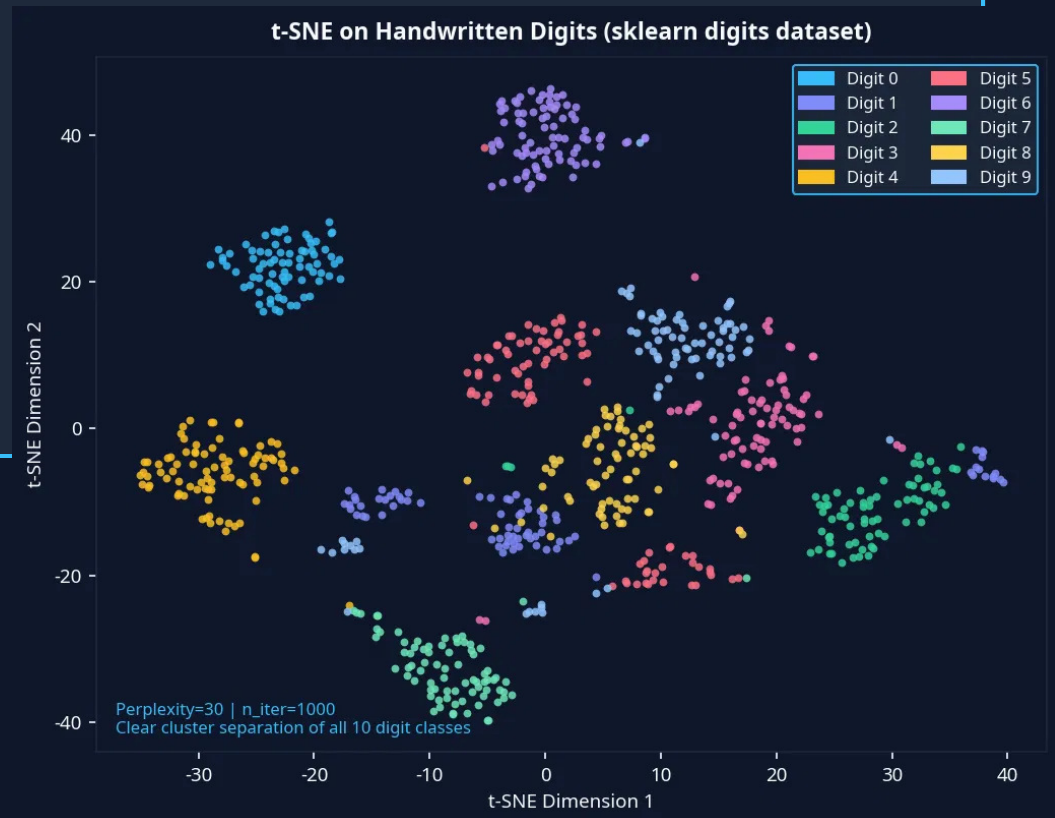
- If p_{ij} is large and q_{ij} is small, the penalty is **huge**.
- If p_{ij} is small and q_{ij} is large, the penalty is **small**.

Conclusion: t-SNE prioritizes preserving local structure over global structure.

GRADIENT DESCENT ANALOGY

The gradient acts like an N-body system:

- **Attractive Force:** Pulls $\mathbf{y}_i$ toward $\mathbf{y}_j$ when p_{ij} is large.
- **Repulsive Force:** Pushes $\mathbf{y}_i$ away when q_{ij} is large.



UMAP: Uniform Manifold Approximation and Projection

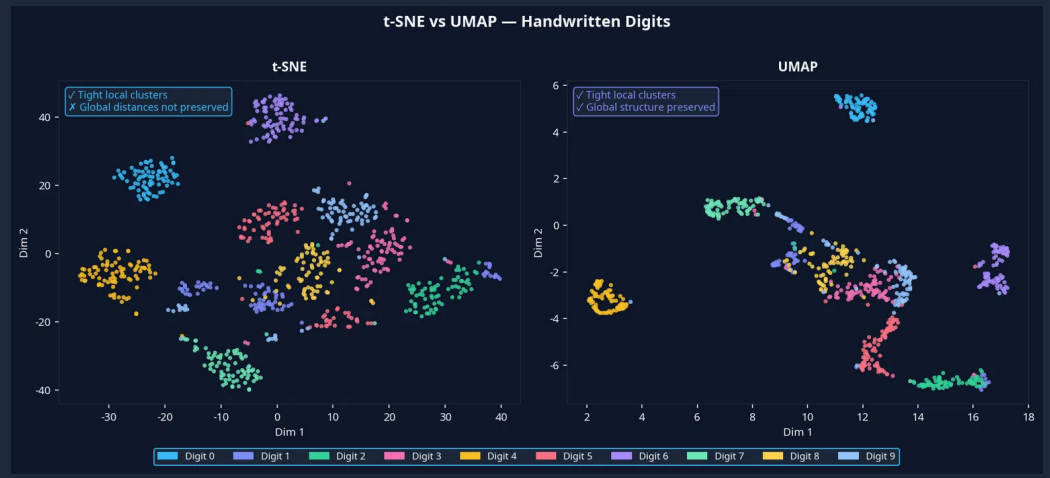
TOPOLOGICAL FOUNDATIONS

UMAP (McInnes et al., 2018) is built on a rigorous mathematical foundation from algebraic topology and Riemannian geometry. It makes three key assumptions:

- The data is uniformly distributed on a Riemannian manifold.
- The Riemannian metric is locally constant (or can be approximated as such).
- The manifold is locally connected.

From these assumptions, UMAP constructs a fuzzy topological representation (a fuzzy simplicial set) of the high-dimensional data.

> **The Goal:** Find a low-dimensional projection of the data that has the closest possible equivalent fuzzy topological structure.



Visual Comparison: While t-SNE creates tight local clusters, it often shatters the global structure. UMAP preserves both local neighborhoods and the broader global relationships between clusters.

UMAP: Optimization Objective

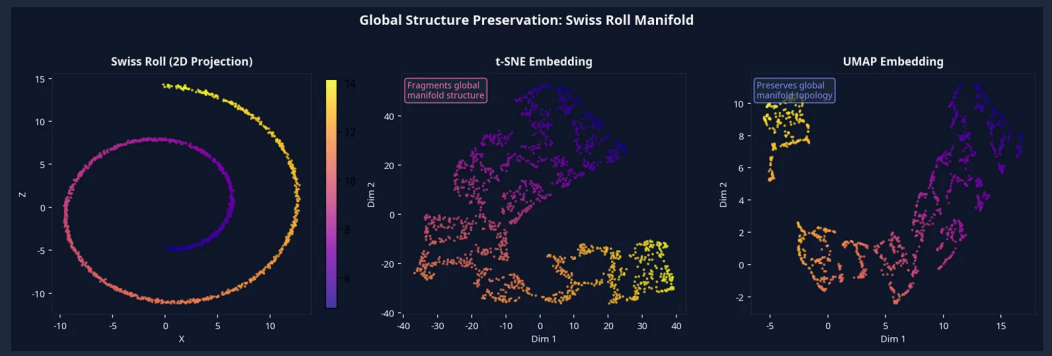
THE CROSS-ENTROPY OBJECTIVE

UMAP minimizes the **cross-entropy** between the high-dimensional fuzzy set P and the low-dimensional fuzzy set Q:

$$C = \sum_{i,j} \left[p_{ij} \log\left(\frac{p_{ij}}{q_{ij}}\right) + (1 - p_{ij}) \log\left(\frac{1 - p_{ij}}{1 - q_{ij}}\right) \right]$$

- **Attractive Term:** $p_{ij} \log(p_{ij}/q_{ij})$. Identical to t-SNE; pulls points together if close in high-D.
- **Repulsive Term:** $(1 - p_{ij}) \log((1 - p_{ij})/(1 - q_{ij}))$. Unique to UMAP; pushes points apart if far in high-D.

Key Insight: t-SNE only models local neighborhoods (attractive forces). UMAP explicitly models both local neighborhoods and global separation (repulsive forces), leading to a more faithful overall representation.



Global Structure Preservation: On a continuous manifold like the Swiss Roll, t-SNE tends to shatter the structure into disconnected fragments. UMAP's repulsive term helps maintain the continuous global topology.

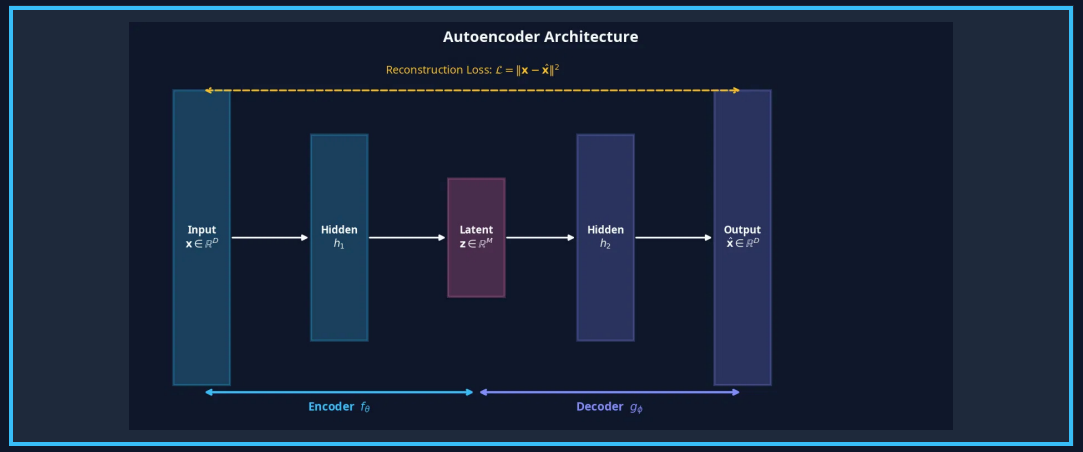
Autoencoders: Neural Dimensionality Reduction

THE ARCHITECTURE

An Autoencoder (AE) copies its input to its output via a bottleneck:

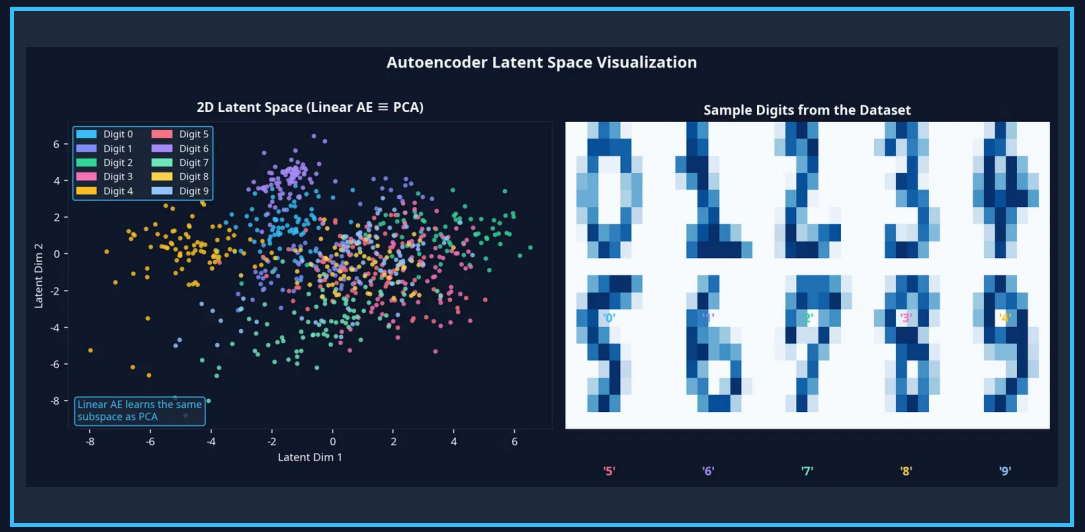
- **Encoder:** Maps input $\mathbf{x} \in \mathbb{R}^D$ to latent $\mathbf{z} \in \mathbb{R}^M$ ($M < D$). $\mathbf{z} = f_\theta(\mathbf{x})$
- **Decoder:** Maps latent $\mathbf{z}$ to reconstruction $\hat{\mathbf{x}} \in \mathbb{R}^D$. $\hat{\mathbf{x}} = g_\phi(\mathbf{z})$

$$\mathcal{L}(\theta, \phi) = \frac{1}{N} \sum_{i=1}^{N} \|\mathbf{x}_i - g_\phi(f_\theta(\mathbf{x}_i))\|^2$$



CONNECTION TO PCA

If f_θ and g_ϕ are strictly **linear** functions, and the loss is MSE, the Autoencoder learns the exact same subspace as PCA.



The Power of Non-Linearity: The true advantage emerges when f_θ and g_ϕ are deep networks with non-linear activations (e.g., ReLU). This allows the AE to learn complex, non-linear manifolds that PCA cannot capture.

Sparse Autoencoders: Inducing Feature Selection

MOTIVATION FOR SPARSITY

Standard autoencoders can learn the identity if the bottleneck is too wide. Even with a narrow bottleneck, learned features are often dense.

Sparse Autoencoders force the network to learn a sparse representation. This leads to:

- **Feature Selection:** Only the most relevant features are active.
- **Disentanglement:** Features become more interpretable.

L1 REGULARIZATION

The simplest way is to add an ℓ_1 penalty on the latent activations $\mathbf{z} = f_\theta(\mathbf{x})$:

$$\mathcal{L}(\theta, \phi) = \frac{1}{N} \sum_{i=1}^{N} \|\mathbf{x}_i - g_\phi(f_\theta(\mathbf{x}_i))\|^2 + \lambda \sum_{j=1}^{k} |z_j|$$

KL DIVERGENCE PENALTY

A common approach is to constrain the average activation of each hidden unit to a small target ρ (e.g. 0.05).

Let $\hat{\rho}_j = \frac{1}{N} \sum_{i=1}^{N} z_{ij}$. We penalize deviations from ρ using the KL divergence:

$$\Omega(\mathbf{z}) = \sum_{j=1}^{k} KL(\rho \| \hat{\rho}_j)$$

$$= \sum_{j=1}^{k} \left[\rho \log \frac{\rho}{\hat{\rho}_j} + (1-\rho) \log \frac{1-\rho}{1-\hat{\rho}_j} \right]$$

The Trade-off: The network balances accurate reconstruction with keeping hidden units inactive, forcing discovery of salient features.

Variational Autoencoders (VAEs): A Generative Approach

MOTIVATION: FROM DETERMINISTIC TO PROBABILISTIC

Standard Autoencoders learn a deterministic mapping so their latent space is unregularized and poor for sampling.

- **The Problem:** Sampling an arbitrary latent z usually yields garbage because the decoder only learned specific training points.
- **The Solution:** VAEs map inputs to a probability distribution, enabling meaningful sampling from the latent space.

THE GENERATIVE MODEL & KEY COMPONENTS

We assume data x is generated from an unobserved latent z. The main pieces are:

- **Prior:** typically an isotropic Gaussian $p(\mathbf{z}) = \mathcal{N}(0, I)$
- **Likelihood (Decoder):** a network parameterizes $p_\theta(\mathbf{x}|\mathbf{z})$
- **Approximate Posterior (Encoder):** we use a tractable q with parameters output by the encoder

$$q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mu_\phi(\mathbf{x}), \mathrm{diag}(\sigma_\phi^2(\mathbf{x})))$$

VAEs: Deriving the Evidence Lower Bound (ELBO)

THE INTRACTABLE INTEGRAL

Our goal is to maximize the marginal likelihood of the data, **p(x)**. However, this requires marginalizing out the latent variable z:

$$p(\mathbf{x}) = \int p_\theta(\mathbf{x}|\mathbf{z})p(\mathbf{z})\,d\mathbf{z}$$

This integral is intractable for complex non-linear models, so the true posterior p(z|x) is also intractable.

THE APPROXIMATE POSTERIOR

We introduce an **inference model** (encoder) **q_φ(z|x)** to approximate the posterior and minimize the KL divergence between them, i.e. *D_{KL}(q_φ||p)*.

DERIVING THE ELBO

Expanding the KL divergence and rearranging terms yields an identity relating log p(x) and an expectation under q_φ. Key steps are compacted here for clarity.

$$\log p(\mathbf{x}) - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})\,||\,p(\mathbf{z}|\mathbf{x})) = \mathbb{E}_{q_\phi}\left[\log p_\theta(\mathbf{x},\mathbf{z}) - \log q_\phi(\mathbf{z}|\mathbf{x})\right]$$

The ELBO Objective: Since $D_{KL} \geq 0$, the right side is a lower bound on log p(x). We maximize the ELBO:

$$\mathcal{L}(\theta,\phi;\mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})\,||\,p(\mathbf{z}))$$

(Reconstruction Likelihood) - (Regularization Penalty)

VAEs: The Reparameterization Trick

THE PROBLEM: STOCHASTIC NODES

To optimize the ELBO using standard backpropagation, we need to compute gradients with respect to the encoder parameters ϕ.

However, the latent variable $\mathbf{z}$ is sampled from the approximate posterior:

$$\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\phi, \boldsymbol{\sigma}_\phi^2)$$

Sampling is a stochastic operation. We cannot backpropagate gradients through a random node because the sampling operation is not differentiable with respect to $\boldsymbol{\mu}_\phi$ and $\boldsymbol{\sigma}_\phi$.

THE SOLUTION: REPARAMETERIZATION

The reparameterization trick expresses the random variable $\mathbf{z}$ as a deterministic function of the parameters ϕ and an independent, parameter-free random noise variable $\boldsymbol{\epsilon}$.

For a Gaussian distribution, we sample $\boldsymbol{\epsilon}$ from a standard normal distribution:

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

And then compute $\mathbf{z}$ deterministically:

$$\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}) + \boldsymbol{\sigma}_\phi(\mathbf{x}) \odot \boldsymbol{\epsilon}$$

Why it works: The stochasticity is now isolated in $\boldsymbol{\epsilon}$, which has no parameters. The path from ϕ to $\mathbf{z}$ is purely deterministic and differentiable, allowing gradients to flow freely back through the network.

Summary: A Taxonomy of Latent Variable Models

LINEAR VS. NON-LINEAR

- **Linear Models:** PCA, CCA, ICA. These models assume the data lies on or near a linear subspace. They are computationally efficient, highly interpretable, and provide exact, closed-form solutions (often via eigendecomposition). Best for simple, global structure.
- **Non-Linear Models:** Kernel PCA, t-SNE, UMAP, Autoencoders. These models can capture complex, curved manifolds. They are more computationally intensive and often rely on iterative optimization (gradient descent). Best for complex, high-dimensional data like images or text.

DETERMINISTIC VS. PROBABILISTIC

- **Deterministic Models:** Standard PCA, Autoencoders, t-SNE, UMAP. These map a data point to a single, fixed point in the latent space. Excellent for visualization, compression, and feature extraction, but poor for generating new data.
- **Probabilistic Models:** Probabilistic PCA, Variational Autoencoders (VAEs). These map data to a probability distribution in the latent space. They provide a rigorous framework for handling uncertainty, missing data, and, crucially, allow for the generation of novel, realistic samples.

> **The Takeaway:** There is no "best" method. The choice depends entirely on the goal: Is it visualization (t-SNE/UMAP), feature selection (Sparse AE), or generation (VAE)?